Proof of Mathar's conjecture on OEIS A027182: a closed form and the six-term recurrence

Haobo Ma* Wenlin Zhang†

1 November 2026

Abstract

We prove the recurrence for A027182 stated as a conjecture by R. J. Mathar (OEIS A027182, FORMULA section, 6 May 2026) for all $n \geq 6$, by evaluating the middle coefficient of the square of each A027170 row polynomial. Row symmetry reduces the convolution to five binomial sums, and a shifted second-order recurrence gives the six-term recurrence. The same central-binomial identity proves Mathar's conjectured recurrences for A027171, A027174 and A027175; a parity decomposition also proves his seven-term recurrence for the half-row sums A027179.

Clark Kimberling's sequence A027182 is defined from the row polynomials of A027170. Mathar conjectured a polynomial-coefficient recurrence for it on 6 May 2026 [1]. We prove that recurrence by first evaluating the convolution explicitly. General algorithms for deriving recurrences of binomial sums are developed by Bostan, Lairez and Salvy [8, Introduction and §4]; here the row symmetry permits a direct evaluation by the binomial theorem. The related closed forms of Christian Krause in A027171, A027174 and A027175 [3, 4, 5] reduce three further conjectures of Mathar, also dated 6 May 2026, to the same central-binomial identity. The half-row conjecture in A027179 [6] follows after separating even and odd rows. We give those deductions below, crediting the recorded closed forms separately from the recurrence proofs.

For integers $n \geq 0$ and $0 \leq k \leq n$, follow A027170 [2] and set

$$\begin{aligned} t(n,k) &= \binom{n+2}{k+1} + \binom{n+1}{k} + \binom{n+1}{k+1} + \binom{n}{k}, \\ T(n,k) &= t(n,k) - t(0,0) + 1 = t(n,k) - 4, \\ g_n(x) &= \sum_{k=0}^{n} T(n,k)x^k, \qquad a_n = [x^n]g_n(x)^2. \end{aligned}$$

Here $[x^n]$ denotes coefficient extraction. The original A027182 indexing starts at $n = 1$; the same definition gives $a_0 = 1$.

Theorem 1. *For every integer $n \geq 0$,*

$$a_n = 4\binom{2n+4}{n+2} + 4\binom{2n+2}{n+1} + \binom{2n}{n} - 72 \cdot 2^n + 16n + 40. \tag{1}$$

*ChronoAI Pte Ltd, Singapore.

†National University of Singapore; ChronoAI Pte Ltd, Singapore.

Consequently, for every integer $n \geq 6$,

$$\begin{aligned} 0 = {} & 2(n+2)a_n - (17n+23)a_{n-1} + (47n+36)a_{n-2} \\ & - 2(22n+17)a_{n-3} + 4(20-n)a_{n-4} \\ & + 8(2n-9)a_{n-5} + 24(12n-47). \end{aligned} \tag{2}$$

Proof. Put $A_k = \binom{n+2}{k+1}$ and $B_k = \binom{n}{k}$. Pascal's identity gives $T(n,k) = 2A_k + B_k - 4$. Since $A_{n-k} = A_k$ and $B_{n-k} = B_k$, the row is symmetric. Hence

$$a_n = \sum_{k=0}^{n} T(n,k)T(n,n-k) = \sum_{k=0}^{n} (2A_k + B_k - 4)^2.$$

The required sums, all with $0 \leq k \leq n$, are

$$\sum A_k^2 = \binom{2n+4}{n+2} - 2, \qquad \sum A_k B_k = \binom{2n+2}{n+1}, \qquad \sum B_k^2 = \binom{2n}{n},$$
$$\sum A_k = 2^{n+2} - 2, \qquad \sum B_k = 2^n.$$

For completeness, coefficient extraction from $(1+x)^r(1+x)^s = (1+x)^{r+s}$ gives $\sum_j \binom{r}{j}\binom{s}{q-j} = \binom{r+s}{q}$. Taking $(r,s,q) = (n+2, n+2, n+2)$ gives the first squared sum after removing its two endpoint terms. Taking $(r,s,q) = (n+2, n, n+1)$ and using $\binom{n}{k} = \binom{n}{n-k}$ gives the mixed sum; its omitted $j=0$ term is zero. Taking $(r,s,q) = (n,n,n)$ gives the remaining squared sum. The two ordinary sums follow by putting $x = 1$ in the binomial theorem, with two endpoint terms removed for A_k. Expanding the square now gives

$$a_n = 4\sum A_k^2 + 4\sum A_k B_k + \sum B_k^2 - 16\sum A_k - 8\sum B_k + 16(n+1),$$

which is exactly (1).

We next derive (2), keeping the variable coefficients inside each shift. Write

$$c_m = \binom{2m}{m}, \qquad h_m = 4c_{m+2} + 4c_{m+1} + c_m, \qquad p_m = 16m + 40.$$

For any sequence f, define

$$(Bf)_m = 2(m+4)f_{m+2} - (7m+27)f_{m+1} - (4m+2)f_m.$$

The factorial definition of c_m implies $R_m := (m+1)c_{m+1} - 2(2m+1)c_m = 0$. Expanding coefficients of $c_m, \ldots, c_{m+4}$ yields

$$(Bh)_m = 8R_{m+3} + 12R_{m+2} + 6R_{m+1} + R_m = 0.$$

Let $(Ef)_m = f_{m+1}$ and $Q(E) = E^3 - 5E^2 + 8E - 4 = (E-1)(E-2)^2$. Define $L = Q(E)B$, meaning that E shifts the entire sequence Bf. Expansion gives

$$\begin{aligned} (Lf)_m = {} & 2(m+7)f_{m+5} - (17m+108)f_{m+4} \\ & + (47m+271)f_{m+3} - (44m+254)f_{m+2} \\ & + (60-4m)f_{m+1} + (16m+8)f_m. \end{aligned} \tag{3}$$

In particular $Lh = 0$. Direct substitution gives

$$(B(2^\bullet))_m = (-10m-24)2^m, \qquad (Bp)_m = -144m^2 - 744m - 1016.$$

The factor $(E-2)^2$ kills every sequence $(\alpha m+\beta)2^m$. For a quadratic $u_m=\alpha m^2+\beta m+\gamma$, expansion gives $(Q(E)u)_m=2\alpha m+\beta-3\alpha$. Consequently,

$$L(2^\bullet)=0, \qquad (Lp)_m=-288m-312=-24(12m+13).$$

Equation (1) says $a_m=h_m-72\cdot 2^m+p_m$. Linearity therefore gives $(La)_m+24(12m+13)=0$ for every $m\ge 0$. Putting $m=n-5$ in (3) proves (2) for $n\ge 6$. □

Initial values and source convention. The definition gives

$$(a_0,a_1,a_2,a_3,a_4)=(1,18,150,820,3726).$$

The example in A027182 prints a final value of 120 for $5^2+10^2+5^2$; the sum is 150, agreeing with the entry's term list and with (1). With the extension $a_0=1$, the same recurrence also holds at $n=5$.

Four companion conjectures

The following corollaries prove the statements labeled "Conjecture" in the FORMULA sections of A027171, A027174, A027175 and A027179, each attributed there to R. J. Mathar on 6 May 2026.

Corollary 2. *Let $u_n=T(2n,n)$ for $n\ge 0$ (A027171), and $v_n=T(2n-1,n-1)$ for $n\ge 1$ (A027175). Then*

$$2(n+1)f_n-(7n+5)f_{n-1}+2(3-2n)f_{n-2}+12(1-3n)=0 \tag{4}$$

holds for $f=u$, $n\ge 2$, and for $f=v$, $n\ge 3$.

Proof. Krause's formulas [3, 5], dated 10 and 11 May 2026, respectively, follow immediately from the definition of T and give

$$u_n=d_n-4, \qquad v_n=\tfrac{1}{2}d_n-4, \qquad d_n=2c_{n+1}+c_n=\frac{9n+5}{n+1}c_n.$$

For the second formula, use $\binom{2n+1}{n}=(2n+1)c_n/(n+1)$ and $\binom{2n-1}{n-1}=c_n/2$. In terms of the already established $R_m=0$, expansion gives

$$2(n+1)d_n-(7n+5)d_{n-1}+2(3-2n)d_{n-2}=4R_n+4R_{n-1}+R_{n-2}=0.$$

The sum of the three coefficients on the left is $3-9n$. Subtracting 4 therefore contributes $36n-12$, which is canceled by $12(1-3n)$. The smallest indices in the statement keep all original sequence indices in their domains. □

Corollary 3. *For $z_n=T(2n,n-2)$, $n\ge 2$ (A027174), and every $n\ge 4$,*

$$\begin{aligned}0=\ &-2(n+3)(139n-175)z_n+(1279n^2+1334n-1089)z_{n-1}\\ &-2(167n+121)(2n-3)z_{n-2}+12(111n^2+456n+229).\end{aligned} \tag{5}$$

Proof. Krause's formula of 11 May 2026 [4], also immediate from T, can be written

$$z_n=2\binom{2n+2}{n-1}+\binom{2n}{n-2}-4=r(n)c_n-4, \qquad r(n)=\frac{n(9n^2+14n+1)}{(n+1)(n+2)(n+3)}.$$

Indeed, the ratios of the binomial coefficients themselves to c_n are $2n(2n+1)/((n+2)(n+3))$ and $n(n-1)/((n+1)(n+2))$; the first has multiplier 2 in z_n. Denote the three coefficients of z_n, z_{n-1}, z_{n-2} in (5) by q_0, q_1, q_2. Since

$$\frac{c_{n-1}}{c_n} = \frac{n}{2(2n-1)}, \qquad \frac{c_{n-2}}{c_n} = \frac{n(n-1)}{4(2n-1)(2n-3)},$$

the homogeneous part for $r(n)c_n$ vanishes by the polynomial identity

$$\begin{aligned} 0 = &-4n(2n-1)(139n-175)(9n^2+14n+1) \\ &+(n-1)(9n^2-4n-4)(1279n^2+1334n-1089) \\ &-(n-2)(n+2)(167n+121)(9n^2-22n+9). \end{aligned} \tag{6}$$

More precisely, its left side is $2(n+1)(n+2)(2n-1)\sum_{j=0}^{2} q_j(n)r(n-j)c_{n-j}/c_n$; expanding the displayed products gives zero. All denominators are nonzero for $n \geq 4$. Finally, $q_0+q_1+q_2 = 3(111n^2+456n+229)$, so the correction -4 supplies exactly the negative of the inhomogeneous term. □

Corollary 4. *For $b_n = \sum_{k=0}^{\lfloor n/2 \rfloor} T(n,k)$, $n \geq 0$ (A027179), put*

$$\begin{aligned} p_0(n) &= -(n+2)(3n-20), & p_1(n) &= -(n+1)(3n-11), \\ p_2(n) &= 33n^2-180n-100, & p_3(n) &= 15n^2-56n+1, \\ p_4(n) &= 2(-51n^2+313n-208), & p_5(n) &= -12(n-1)(n-4), \\ p_6(n) &= 8(9n-11)(n-5). \end{aligned}$$

Then Mathar's seven-term recurrence holds for every $n \geq 6$:

$$\sum_{j=0}^{6} p_j(n) b_{n-j} + 4(-24n^2+196n-365) = 0. \tag{7}$$

Proof. Summing the two binomial terms in $T(n,k)$ gives

$$S_n := \sum_{k=0}^{n} T(n,k) = 2(2^{n+2}-2) + 2^n - 4(n+1) = 9\cdot 2^n - 4n - 8.$$

This is the row-sum formula recorded by Ralf Stephan in 2004 and proved by Krause in 2026 in A027178 [7]. By row symmetry, the half-row sum is $S_n/2$ when n is odd and $(S_n + T(n, n/2))/2$ when n is even, as recorded by Mathar [6]. Define

$$w_{2m} = d_m = \frac{9m+5}{m+1} c_m, \qquad w_{2m+1} = 0 \quad (m \geq 0).$$

In even rows, $T(2m,m) = d_m - 4$. Since the indicator of even n is $(1+(-1)^n)/2$, this gives the all-index closed form

$$b_n = \frac{9}{2}\, 2^n - 2n - 5 - (-1)^n + \frac{1}{2} w_n \qquad (n \geq 0). \tag{8}$$

Write $(\mathcal{P}f)_n = \sum_{j=0}^{6} p_j(n) f_{n-j}$. Expanding the seven coefficients gives

$$\sum_{j=0}^{6} p_j(n) 2^{-j} = 0, \qquad \sum_{j=0}^{6} p_j(n)(-1)^j = 0,$$

$$\sum_{j=0}^{6} p_j(n)(-2(n-j)-5) = 4(24n^2-196n+365).$$

Thus it remains to check $\mathcal{P}w = 0$. For even $n \geq 6$, only shifts $j = 0, 2, 4, 6$ contribute. Dividing by $c_{n/2}$ and using the ratio for consecutive central binomial coefficients gives

$$\frac{(\mathcal{P}w)_n}{c_{n/2}} = \frac{E(n)}{8(n-3)(n-1)},$$

where the explicit certificate is

$$\begin{aligned} E(n) = & -8(n-3)(n-1)(3n-20)(9n+10) \\ & +2(n-3)(9n-8)(33n^2-180n-100) \\ & -n(9n-26)(51n^2-313n+208) \\ & +n(n-2)(9n-44)(9n-11) = 0. \end{aligned} \tag{9}$$

For odd $n \geq 7$, only shifts $j = 1, 3, 5$ contribute, and similarly

$$\frac{(\mathcal{P}w)_n}{c_{(n-1)/2}} = \frac{O(n)}{4(n-2)},$$

with

$$\begin{aligned} O(n) = & -4(n-2)(3n-11)(9n+1) \\ & +(9n-17)(15n^2-56n+1) - 3(n-1)^2(9n-35) = 0. \end{aligned} \tag{10}$$

Both final equalities follow by expanding the displayed products. For clarity, all terms before clearing denominators can be recovered from

$$\frac{w_{n-j}}{c_{(n-\epsilon)/2}} = \frac{9(n-j)+10}{n-j+2} \prod_{\ell=0}^{(j-\epsilon)/2-1} \frac{n-\epsilon-2\ell}{4(n-\epsilon-2\ell-1)}, \quad j \equiv \epsilon \equiv n \pmod 2,$$

where an empty product is 1. The other shifts give zero. There is no zero denominator in the stated ranges. Applying $\mathcal{P}$ to (8) now proves (7). □

Verification and availability. The argument above proves the identities for all stated indices. The accompanying Python check separately compares the defining convolution with (1) through $n = 400$, checks all five recurrences through $n = 400$, and checks the displayed polynomial certificates exactly. These finite checks are separate from the all-index proofs above. The pinned file `MatharA027182SelfConvolution.lean` (commit `494f471b70d9`) has the same $n \geq 6$ statement as theorem `MatharA027182SelfConvolution.result`. The companion corollaries have the written proofs and exact arithmetic checks given here.

License. This note is licensed under the Creative Commons Attribution 4.0 International license (CC BY 4.0).

References

[1] C. Kimberling, *A027182: a(n) = self-convolution of row n of array T given by A027170*, The On-Line Encyclopedia of Integer Sequences, FORMULA section: R. J. Mathar's conjecture of 6 May 2026. `https://oeis.org/A027182`. Accessed 1 November 2026.

[2] C. Kimberling, *A027170: Triangular array T read by rows (4-diamondization of Pascal's triangle)*, The On-Line Encyclopedia of Integer Sequences. https://oeis.org/A027170. Accessed 1 November 2026.

[3] C. Kimberling, *A027171: a(n) = A027170(2n, n)*, The On-Line Encyclopedia of Integer Sequences, FORMULA section: Mathar's conjecture of 6 May 2026 and Krause's closed form of 10 May 2026. https://oeis.org/A027171. Accessed 1 November 2026.

[4] C. Kimberling, *A027174: a(n) = A027170(2n, n-2)*, The On-Line Encyclopedia of Integer Sequences, FORMULA section: Mathar's conjecture of 6 May 2026 and Krause's closed form of 11 May 2026. https://oeis.org/A027174. Accessed 1 November 2026.

[5] C. Kimberling, *A027175: a(n) = A027170(2n-1, n-1)*, The On-Line Encyclopedia of Integer Sequences, FORMULA section: Mathar's conjecture of 6 May 2026 and Krause's closed form of 11 May 2026. https://oeis.org/A027175. Accessed 1 November 2026.

[6] C. Kimberling, *A027179: a(n) = T(n,0) + T(n,1) + ... + T(n,[n/2]), T given by A027170*, The On-Line Encyclopedia of Integer Sequences, FORMULA section: Mathar's conjecture and parity formulas of 6 May 2026. https://oeis.org/A027179. Accessed 1 November 2026.

[7] C. Kimberling, *A027178: a(n) = T(n,0) + T(n,1) + ... + T(n,n), T given by A027170*, The On-Line Encyclopedia of Integer Sequences, FORMULA section: Stephan's formula of 13 February 2004, with Krause's proof attribution of 14 May 2026. https://oeis.org/A027178. Accessed 1 November 2026.

[8] A. Bostan, P. Lairez and B. Salvy, *Multiple binomial sums*, Journal of Symbolic Computation **80**, part 2 (2017), 351–386. doi:10.1016/j.jsc.2016.04.002. Preprint: arXiv:1510.07487v2.